THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 18	There are $2n = 18$ numbers between n^2 and $(n+1)^2$
2	**(a)** 8	$\frac{1}{a^{-n}} = a^n = 8$
3	**(c)** IC	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
4	**(b)** 144°	One part $= 360 \div 10 = 36^\circ$
5	**(b)** 3465	alternate digit sums 9 and 9 differ by 0, a multiple of 11
6	**(c)** $x^2 + 6x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
7	**(b)** 4 : 1	Convert to the same unit
8	**(b)** 25%	Change/original × 100 = 20/80×100
9	**(b)** $\sqrt{74}$	$h^2 = 5^2 + 7^2 = 74$
10	**(a)** 8:12:15	Make B equal: B $= \mathrm{LCM}(3,4) = 12$; A $= 8$, C $= 15$ gives $8 : 12 : 15$
11	**(d)** 8	$F = 2 + E - V = 8$
12	**(d)** 16	Even number of values: mean of the two middle values
13	**(a)** $5n$	$5(n+3) - 15 = 5n$
14	**(b)** 15	$b = A/l = 360/24$
15	**(d)** 29	$(n+1)^2 - n^2 = 2n + 1 = 29$
16	**(a)** 81	Even power gives a positive value; odd power gives a negative value
17	**(a)** 73	Multiply each digit by the power of the base and add
18	**(c)** 30	Opposite angles are equal, so the two expressions are equal
19	**(a)** 4518	even and digit sum 18 is a multiple of 3
20	**(a)** $x^2 - 4x + 4$	$(a-b)^2 = a^2 - 2ab + b^2$
21	**(a)** 4 : 6 : 5	Make the middle terms equal
22	**(a)** 25%	18/72×100
23	**(d)** 130	Square on hypotenuse $= a^2 + b^2 = 130$
24	**(a)** 15	$18 \times 10 = 12 \times x$
25	**(c)** 4	Each segment becomes 4: $4^1 = 4$
26	**(a)** 20	Total − known sum
27	**(c)** $3n + 3$	$n + (n+1) + \cdots = 3n + 3$
28	**(c)** 9	$h = 2A/b = 2 \times 72/16$
29	**(d)** 36	$T_n = n(n+1)/2 = 36$
30	**(b)** 6×10^7	Place the decimal point after the first non-zero digit
31	**(c)** 322	Divide repeatedly by the base and read the remainders upwards
32	**(c)** 35°	A diagonal of a rhombus bisects the angle at the vertex
33	**(a)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract
34	**(b)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
35	**(d)** 17.5 km	3.5×5
36	**(a)** 20%	(SP−CP)/CP×100 = 300/1500×100
37	**(c)** 16, 63, 65	$2m = 16$, $m^2 - 1 = 63$, $m^2 + 1 = 65$
38	**(b)** 9	$135 \div 15$
39	**(d)** $\frac{9}{16}$	Each stage keeps $\frac{3}{4}$: $(3/4)^2$
40	**(a)** 22	New sum 132, new count 6

41	**(b)** 20	Top $= a + 3b + 3c + d$
42	**(c)** 100 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 20 \times (6 + 4)$
43	**(b)** 49	$T_n + T_{n+1} = (n+1)^2 = 49$
44	**(c)** 0.00407	Move the decimal point according to the sign of the exponent
45	**(c)** 135	Base 60: value = 60 x first + second
46	**(a)** 50°	OA = OB, so triangle OAB is isosceles
47	**(a)** 39	Sum $= 3\times$ middle number
48	**(c)** $x^2 + 7x + 10$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
49	**(b)** 10 days	Fewer or more workers: days vary inversely
50	**(c)** 20%	Discount = 500; 500/2500×100

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** right-angled	$14^2 + 48^2 = 2500$ and $50^2 = 2500$
2	**(b)** 84	$14/60 \times 360$
3	**(c)** 6	$6(n-2)^2 = 6 \times 1$
4	**(a)** Between week 2 and week 3	Growth in each week: 4, 7, 4, 3, 6; largest is 7
5	**(d)** 90	Sum $= 9\times$ middle date
6	**(b)** 9720	$18 \times 12 \times 45$
7	**(c)** 8, 15, 17	$2m = 8$, $m^2 - 1 = 15$, $m^2 + 1 = 17$
8	**(a)** 2^{-6}	$\frac{1}{2^6} = 2^{-6}$
9	**(c)** 72	Base 20: value = 20 x first + second
10	**(a)** 110°	Angle sum is 360°
11	**(d)** 80	Sum $= 5\times$ middle number
12	**(b)** 9999	$(100+1)(100-1) = 100^2 - 1^2$
13	**(a)** 420 km	Speed $240 \div 4$
14	**(b)** 2100	GST = 100; bill = 2100
15	**(d)** $3\sqrt{2}$	$d^2 = s^2 + s^2 = 18$
16	**(c)** 900	One part $=$ ₹300
17	**(a)** two circles and one rectangle	Unrolling the curved surface gives a rectangle
18	**(d)** 7	The value with most dots is 7
19	**(a)** 1	$n^2 - (n-1)(n+1) = 1$
20	**(d)** 20000	$1\ \mathrm{m}^2 = 10000\ \mathrm{cm}^2$
21	**(d)** 100	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 100$
22	**(c)** 7	The number being multiplied repeatedly is the base
23	**(c)** Hindu-Arabic system	Digits 0-9 with place value came from India and spread through Arabs
24	**(b)** 115°	Angles on the same non-parallel side add up to 180°
25	**(c)** 14	$(7+5) \times 2 - 10 = 2 \times 7$
26	**(b)** 22	Add the two cross products
27	**(c)** ₹ 150	One part $= 150$
28	**(a)** 1050	Amount = 5000(1+10/100)² = 6050; CI = A − P
29	**(d)** 210	Other leg $= 21$ cm; area $= \frac{1}{2} \times 20 \times 21$
30	**(a)** 560	Speed $= 70$ km/h
31	**(d)** circle (with a dot at the centre)	Looking down, the base circle with the apex at centre
32	**(a)** 6	$\sum xf = 72$, $\sum f = 12$
33	**(b)** $4n$	Expand: $4an = 4 \times 1n$
34	**(b)** 8	$a + b = 2A/h = 20$
35	**(a)** 24	There are $2n = 24$ numbers between n^2 and $(n+1)^2$
36	**(b)** 5	One lakh is 10^5
37	**(a)** VX	I can be subtracted only from V and X; X only from L and C; C only from D and M; V, L, D are never subtracted
38	**(c)** The diagonals bisect each other	Parallelogram diagonals bisect each other
39	**(d)** 1	Check the rule for 9
40	**(b)** $(a-b)^2 = a^2 - 2ab + b^2$	Check by expanding

41	**(c)** decreases	Inverse proportion
42	**(a)** 20%	Change/original × 100 = 50/250×100
43	**(d)** the square of the hypotenuse equals the sum of the squares of the other two sides	Statement of the theorem
44	**(c)** 5:8	Compare as fractions: $\frac{5}{8} \approx 0.625$, $\frac{7}{12} \approx 0.583$, $\frac{3}{5} \approx 0.600$, $\frac{4}{7} \approx 0.571$
45	**(b)** 7	$F = 2 + E - V = 7$
46	**(c)** 19	Even number of values: mean of the two middle values
47	**(a)** $4n$	$4(n+6) - 24 = 4n$
48	**(a)** 11	$b = A/l = 352/32$
49	**(a)** 39	$(n+1)^2 - n^2 = 2n + 1 = 39$
50	**(c)** 9	$\frac{1}{a^{-n}} = a^n = 9$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 125	Multiply each digit by the power of the base and add
2	**(b)** 144°	One part $= 360 \div 15 = 24^\circ$
3	**(a)** odd	Odd + odd = even, even + odd = odd
4	**(d)** $x^2 + 8x + 16$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(a)** 5 : 1	Convert to the same unit
6	**(b)** 25%	35/140×100
7	**(c)** $2\sqrt{10}$	$h^2 = 2^2 + 6^2 = 40$
8	**(d)** 15:20:24	Make B equal: B $= \mathrm{LCM}(4,5) = 20$; A $= 15$, C $= 24$ gives $15 : 20 : 24$
9	**(b)** 16	Each segment becomes 4: $4^2 = 16$
10	**(c)** 26	Total − known sum
11	**(d)** $4n + 6$	$n + (n+1) + \cdots = 4n + 6$
12	**(a)** 13	$h = 2A/b = 2 \times 130/20$
13	**(b)** 55	$T_n = n(n+1)/2 = 55$
14	**(d)** -32	Even power gives a positive value; odd power gives a negative value
15	**(c)** 226	Divide repeatedly by the base and read the remainders upwards
16	**(d)** 15	Opposite angles are equal, so the two expressions are equal
17	**(c)** 8239	alternate digit sums 11 and 11 differ by 0, a multiple of 11
18	**(b)** $x^2 - 10x + 25$	$(a-b)^2 = a^2 - 2ab + b^2$
19	**(c)** 6 : 8 : 7	Make the middle terms equal
20	**(b)** 10%	(SP−CP)/CP×100 = 240/2400×100
21	**(b)** 290	Square on hypotenuse $= a^2 + b^2 = 290$
22	**(a)** 12	$20 \times 9 = 15 \times x$
23	**(d)** $\frac{27}{64}$	Each stage keeps $\frac{3}{4}$: $(3/4)^3$
24	**(c)** 30	New sum 180, new count 6
25	**(d)** 23	Top $= a + 3b + 3c + d$
26	**(c)** 144 cm²	$\frac{1}{2}d(h_1 + h_2) = \frac{1}{2} \times 24 \times (5+7)$
27	**(a)** 100	$T_n + T_{n+1} = (n+1)^2 = 100$
28	**(d)** 3.844×10^5	Place the decimal point after the first non-zero digit
29	**(d)** 220	Base 60: value = 60 x first + second
30	**(d)** 50°	A diagonal of a rhombus bisects the angle at the vertex
31	**(b)** 6150	even and digit sum 12 is a multiple of 3
32	**(d)** $x^2 - 36$	$(a+b)(a-b) = a^2 - b^2$
33	**(b)** 48 km	4×12
34	**(c)** 10%	Discount = 180; 180/1800×100
35	**(b)** 18, 80, 82	$2m = 18$, $m^2 - 1 = 80$, $m^2 + 1 = 82$
36	**(b)** 7	$175 \div 25$
37	**(a)** 24	$6(n-2)^2 = 6 \times 4$
38	**(a)** Between week 3 and week 4	Growth in each week: 3, 2, 6, 0, 0; largest is 6
39	**(b)** 108	Sum $= 9\times$ middle date
40	**(c)** 15000	$25 \times 10 \times 60$
41	**(b)** 10, 24, 26	$2m = 10$, $m^2 - 1 = 24$, $m^2 + 1 = 26$
42	**(b)** 56200	Move the decimal point according to the sign of the exponent

43	**(b)** 107	Base 20: value = 20 x first + second
44	**(b)** 40°	OA = OB, so triangle OAB is isosceles
45	**(c)** 6174	Kaprekar routine: arrange digits in descending and ascending order and subtract
46	**(b)** $x^2 + 10x + 21$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
47	**(d)** 15 days	Fewer or more workers: days vary inversely
48	**(b)** 1680	GST = 180; bill = 1680
49	**(a)** not right-angled	$11^2 + 25^2 = 746$ and $30^2 = 900$
50	**(b)** 144	$18/45 \times 360$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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